Probationary period 1

Consolidated report

Development of an LLM Agent for Processing and Qualifying Leads from a Messaging Platform with CRM Integration

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Date: 24 October 2026
Repository version: tag prob-period-1

Repository: https://github.com/Wolfenste1n/diploma-Madiyar
Tagged version: https://github.com/Wolfenste1n/diploma-Madiyar/tree/prob-period-1

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Links to the repository version tagged prob-period-1

All materials in this report match the repository version tagged **prob-period-1**. The repository is public, so every link below opens without a login.

Material	Link
Repository	https://github.com/Wolfenste1n/diploma-Madiyar
Tagged version (all files)	https://github.com/Wolfenste1n/diploma-Madiyar/tree/prob-period-1
Documentation, sections 1 to 11 (Markdown)	https://github.com/Wolfenste1n/diploma-Madiyar/tree/prob-period-1/docs
Prototype (mockups of key scenarios)	https://github.com/Wolfenste1n/diploma-Madiyar/tree/prob-period-1/prototype
Editable diagram sources (Mermaid .mmd)	https://github.com/Wolfenste1n/diploma-Madiyar/tree/prob-period-1/diagrams
System context diagram source	https://github.com/Wolfenste1n/diploma-Madiyar/blob/prob-period-1/diagrams/system-context.mmd
Use case diagram source	https://github.com/Wolfenste1n/diploma-Madiyar/blob/prob-period-1/diagrams/use-case.mmd
Component diagram source	https://github.com/Wolfenste1n/diploma-Madiyar/blob/prob-period-1/diagrams/component.mmd
Diagram exports (PNG)	https://github.com/Wolfenste1n/diploma-Madiyar/tree/prob-period-1/diagrams/png
Source papers (PDF) and index	https://github.com/Wolfenste1n/diploma-Madiyar/tree/prob-period-1/papers
Sources table (Excel)	https://github.com/Wolfenste1n/diploma-Madiyar/blob/prob-period-1/papers/Diploma_sources.xlsx
Task plan (PDF)	https://github.com/Wolfenste1n/diploma-Madiyar/blob/prob-period-1/docs/DIP_task_plan.pdf
YouTrack project DIP	https://madiyar-diploma.youtrack.cloud/dashboard?id=177-0

Sections: 1 Project proposal, 2 Evidence of relevance, 3 Literature review, 4 Existing system comparison, 5 Technology comparison, 6 Requirements specification, 7 Architecture and diagrams, 8 UX / prototype, 9 Verification and evaluation plan, 10 Project management and risks, 11 Supervisor feedback.

1. Project proposal

Project title. Development of an LLM Agent for Processing and Qualifying Leads from a Messaging Platform with CRM Integration

- Russian: Разработка LLM-агента для обработки и квалификации лидов из мессенджера с интеграцией в CRM-систему
- Kazakh: Мессенджер арқылы келіп түскен лидтерді өңдеу және саралау үшін CRM жүйесімен біріктірілген LLM-агентін әзірлеу

Team. Individual project. Student: Madiyar Kenzhebayev, group SE-2407, Astana IT University (AITU).

Supervisor. Zhanna B. Sadirmekova, PhD, Associate Professor.

Topic status. Approved by the supervisor; committee approval pending.

1.1 Problem statement

Small businesses receive leads through messengers (Telegram, WhatsApp) in free text and at any hour. A manager reads each message, asks follow-up questions (what product, what size, how urgent, delivery or pickup) and copies the result into a CRM by hand. When the number of messages exceeds the manager's capacity, leads wait for a reply and CRM records are incomplete or missing. Rule-based chatbots cover only fixed scripts and fail on free-form questions. Commercial AI agents exist, but none of the reviewed products publishes a method for measuring the quality of lead qualification (section 4).

1.2 Target users

User	Need	Interaction with the system
Customer (lead)	A fast, relevant answer and a clear next step	Writes in Telegram or WhatsApp
Sales manager	Complete, prioritized lead cards; a handoff when a human is required	Works in the CRM (Bitrix24)
Operator (LeadUp agency)	Control of questions, rules and quality	Configures rules and reads metrics

Pilot domain: plumbing and heating supplies, Altay Santeh Opt (Astana).

1.3 Aim

Build and evaluate an LLM agent that qualifies leads from messengers and writes validated lead cards to a CRM, handing over to a human when needed.

1.4 Objectives

1. Collect evidence about the process and requirements from the client and domain experts.
2. Review the literature and existing systems; justify technology choices.
3. Specify requirements with acceptance criteria; design the architecture.
4. Implement an MVP: Telegram channel, agent core, CRM adapter (Bitrix24 and a mock CRM), human handoff.
5. Add the WhatsApp channel (Cloud API).
6. Evaluate the agent against baselines on at least 100 labeled dialogues and report the metrics.

1.5 Expected outcomes

- A working MVP (Telegram, Bitrix24 adapter, handoff).
- An evaluation report with metrics compared with baselines.
- Documentation and a reproducible setup (this repository).

1.6 Project scope

In scope	Out of scope (this release)	Stretch goals
Telegram and WhatsApp, text dialogues (Russian first), one CRM adapter (Bitrix24), human handoff, evaluation	Instagram, TikTok, voice, building an own CRM, fine-tuning foundation models	Photo hints, follow-up reminders, Kazakh dialogues

1.7 Engineering complexity and realism of the scope

The engineering complexity lies in stateful dialogue management (an explicit state machine around the LLM), schema-constrained extraction with validation, tool use against a CRM through an adapter, guardrails against hallucination and prompt injection, asynchronous webhook processing and an evaluation harness with baselines. The scope is realistic for one student because the work is MVP-first: one pilot business, one CRM, one channel first (Telegram), and the remaining channels and features are separated as later iterations or stretch goals.

2. Evidence of relevance

2.1 Approach

Evidence was collected through discussions with domain experts, including the pilot client, and through observation of an existing process (the client's content brief). The target is 3 to 5 discussions with user representatives or a substantive agreement with the actual client. Current status: 2 expert discussions held; requirements were approved by the clients.

2.2 Expert discussions

Expert	Role	Company	What the discussion confirmed
Altay	Founder	Altay Santeh Opt (pilot client, Astana)	Manager overloaded: many requests, slow replies
Nikolai Salo	Founder	Salo Marketing (US home-services marketing)	The same overload at US home-services clients

2.3 Observed process and available data

The client's content brief (shooting checklist for nine Reels) shows how leads reach the business: each Reel ends with a call to write a keyword in Direct or WhatsApp (ПОДБОР, КОМПЛЕКТ, КАРТРИДЖ, ФИТИНГ, РАДИАТОР, ПОДКЛЮЧЕНИЕ) and to send a photo and sizes. Inbound chats are therefore expected to be short, free-text and to miss technical details, which the agent must ask for.

2.4 Findings and their status

Finding	Status	Informs requirement
Reels ask for a keyword, photo and sizes in Direct	Verified (client content brief)	FR-04, FR-13
One manager, many requests, not all answered in time	Reported by Altay	FR-07, FR-08, NFR-01
The same overload at US home-services clients	Reported by Nikolai Salo	NFR-08
A faster first reply raises conversion	Assumption	Measured against the human baseline (section 9)

Status legend: **Verified** means confirmed by a document or data; **Reported** means stated in a discussion and not yet measured; **Assumption** means to be tested in the evaluation.

2.5 Evidence still to collect

- Metrics from the client: inbound chats per day, median first-reply time, share of chats without a CRM record.
- An additional discussion with a sales manager (a direct user of the lead cards).
- Anonymized real dialogues for the test set, with written consent (section 9).

3. Literature review

3.1 Selection of sources

Sources were chosen around five themes that the project depends on: LLM agents and tool use, task-oriented dialogue with LLMs, reliability and security (hallucination, prompt injection, output formats), evaluation methods, and lead scoring. The review uses 18 sources: 13 scientific publications (12 from 2023 to 2025 and one foundational work from 2020) and 5 technical documents. Every scientific source has a DOI or arXiv identifier, and copies are stored in the repository (papers folder).

3.2 Structured analysis table

#	Source (venue)	Year	What it shows	Limitation for this project	Identifier
1	Yao et al., ReAct (ICLR)	2023	Reasoning interleaved with tool calls	General benchmarks, not sales chats	arXiv:2210.03629
2	Schick et al., Toolformer (NeurIPS)	2023	Language models learn to call APIs	Training-based; the project uses prompts	arXiv:2302.04761
3	Wang et al., survey of LLM-based agents (Frontiers of Computer Science)	2024	Agent modules: memory, planning, action	Survey; no small-business guidance	arXiv:2308.11432
4	Gao et al., RAG survey (arXiv)	2023	Grounding reduces fabrication	Generic; a domain knowledge base is needed	arXiv:2312.10997
5	Hudeček, Dušek (SIGDIAL)	2023	LLMs are weak at slot tracking	Benchmark dialogues in English	doi:10.18653/v1/2023.sigdial-1.21
6	Yao et al., tau-bench (arXiv)	2024	Judge the end state of tool use	Retail and airline domains	arXiv:2406.12045
7	Ji et al., hallucination survey (ACM Computing Surveys)	2023	Hallucination types and mitigations	Not specific to sales chats	arXiv:2202.03629
8	Zheng et al., LLM-as-a-Judge (NeurIPS D&B)	2023	LLM judges agree with humans	Biased; used only as a secondary scorer	arXiv:2306.05685
9	Greshake et al., indirect prompt injection (AISec)	2023	Untrusted text hijacks tools	Shows attacks; defenses are tested here	arXiv:2302.12173
10	Tam et al., format restrictions (EMNLP Industry)	2024	Strict output formats can hurt reasoning	Effect depends on the model; ablation planned	arXiv:2408.02442
11	González-Flores et al., B2B lead scoring (Frontiers in AI)	2025	Machine learning scores B2B leads	Structured data, not chats	doi:10.3389/frai.2025.1554325
12	Wu, Andreev, Benyoucef, lead scoring review (Information Technology and Management)	2023	Traditional vs predictive scoring; 44 studies	Review; no LLM agents	doi:10.1007/s10799-023-00388-w
13	Lewis et al., RAG (NeurIPS), foundational	2020	Retrieval grounding for NLP	Pre-dates LLM agents	arXiv:2005.11401

3.3 Synthesis

- **What approaches exist.** Tool-calling agents (sources 1 to 3), retrieval grounding (4, 13), explicit evaluation of agent behavior (6, 8) and lead scoring with machine learning on structured data (11, 12).
- **What is already known.** The agent loop is established; reliability is the open issue. LLMs are weak at exact slot tracking (5), hallucinate (7), can be hijacked through untrusted text (9) and can lose reasoning quality under strict output formats (10).
- **Limitations the project takes into account.** The design therefore uses an explicit state machine, schema validation, knowledge-base-only answers for facts, a tool whitelist, end-state evaluation of the CRM record (6) and human labels as the primary metric, with LLM judges only as a secondary scorer.

3.4 Gap

Lead scoring studies work with structured data and do not cover free-text messenger conversations; agent studies use general benchmarks. No evaluated pipeline turns messenger chats into validated CRM lead cards with human handoff for small businesses. This is the gap the project addresses.

3.5 Technical documentation

#	Source	Link
14	HighLevel Support, How to set up a Conversation AI bot	help.gohighlevel.com article
15	Meta, WhatsApp Business Platform, Cloud API	developers.facebook.com
16	Telegram, Bot API	core.telegram.org
17	Bitrix24, REST API	apidocs.bitrix24.com
18	OWASP, Top 10 for LLM Applications	owasp.org

3.6 Supporting materials

- Sources table with authors, venues, identifiers and links: Diploma_sources.xlsx
- Copies of the papers (PDF) and a numbered index: papers/

4. Existing system comparison

4.1 Systems and criteria

Four existing approaches were compared against this project on eight criteria. The set of systems covers an all-in-one CRM platform, a CRM marketplace app, an independent AI-agent service and a do-it-yourself pipeline, which are the options a small business realistically has.

- **HighLevel Conversation AI:** a bot built into a CRM and marketing platform.
- **Bitrix24 with an AI app from the marketplace:** messaging channels plus a third-party AI app.
- **Moon AI:** a third-party AI-agent service connected to Bitrix24 and other CRMs.
- **Make / n8n with an LLM:** a do-it-yourself pipeline (the approach the author used before).

Criterion	HighLevel Conversation AI	Bitrix24 + AI app	Moon AI	Make / n8n + LLM	This project
Channels	WhatsApp, Instagram, SMS, chat	Telegram, WhatsApp, chat	WhatsApp, Telegram, Instagram	Any, built by hand	Telegram, WhatsApp
CRM integration	Native (same platform)	Writes Bitrix24 fields	Lead and deal (vendor claim)	Via API modules	Bitrix24 adapter and mock
Structured fields, validation	Not confirmed	Fields, validation not stated	Filled card (vendor claim)	Custom, none by default	Typed schema, validated
Human handoff	Message limits, modes	To a live manager (listing)	Not confirmed	Custom	Explicit triggers
Published quality method	Not found	Not found	Not found	None	Metrics against baselines
Control of data and prompts	Closed platform	Closed app	Closed service	Medium	Full, own server
Pricing model	Usage-based	License and message units	Subscription and tokens	Platform and API	Tokens and own server
Languages	Not confirmed	Not confirmed	RU, KZ, EN (vendor claim)	Depends on the LLM	Russian first
Evidence source	Help center (official)	Marketplace listings	Product pages (vendor claims)	Author experience	Planned, measured

4.2 Evidence and its limits

- HighLevel: official help article, How to set up a Conversation AI bot.
- Bitrix24: marketplace listings of AI apps (vendor descriptions).
- Moon AI: product pages (vendor claims, not independently verified).
- Make / n8n: the author's own experience with such pipelines.
- "Not confirmed" and "Not found" mean that the reviewed material does not state the feature; they do not prove that it is absent.

4.3 Findings: why a new solution is needed

1. The function itself (qualify a lead and write it to a CRM) is already offered commercially, so the problem is real.
2. None of the reviewed products publishes a method for measuring quality (accuracy of qualification, share of invented facts, correctness of handoff); buyers have to trust claims.
3. Closed platforms limit control over prompts and data; a do-it-yourself pipeline gives control but has no validation or test framework.
4. This project adds a typed lead schema with validation, explicit handoff rules, a replaceable channel, LLM and CRM, and a reproducible evaluation against baselines.

5. Technology comparison

5.1 Key technology decisions

Each decision lists the alternatives that were considered, the choice and the requirement that justifies it. Choices are provisional until the evaluation (section 9) measures quality and cost.

Decision	Alternatives considered	Chosen	Justification (requirements)
LLM access	Hosted API; open-weight model on own GPU; hybrid	Hosted API behind an interface	Best Russian quality for the MVP; replaceable (NFR-08); an open-weight model is compared in the evaluation
Orchestration	Own state machine; LangGraph; Make / n8n	Own state machine	The qualification process is bounded with explicit states; deterministic and testable (FR-04, FR-05)
Server platform	Python with FastAPI; Node.js	Python with FastAPI	LLM and evaluation ecosystem; typed schemas
Data storage	PostgreSQL; MongoDB; SQLite	PostgreSQL	Transactions for state, logs and the knowledge base in one database
WhatsApp access	Cloud API; aggregator; unofficial library	Cloud API (official)	Compliance with platform rules; unofficial libraries risk account bans
CRM	Bitrix24; amoCRM / Kommo; own CRM	Bitrix24 through an adapter	Part of the LeadUp stack; documented REST API; an own CRM is outside the aim
Component interaction	Webhooks with a queue; message broker; synchronous calls	Webhooks with a Redis queue	Fast webhook reply and no lost messages (NFR-01, NFR-02)

Telegram uses the official Bot API.

5.2 Conclusion on technologies

Every choice is justified by a requirement and is replaceable behind an interface (LLM client, channel adapters, CRM adapter), so the contribution stays in the agent logic and its evaluation. Open questions: whether an open-weight model is competitive in Russian and Kazakh (measured in the evaluation) and whether fine-tuning is needed (only if the supervisor requires it).

6. Requirements specification

Requirements were approved by the clients. Priorities follow MoSCoW. Numeric thresholds are initial targets and will be calibrated after the baseline is measured (section 9).

6.1 Functional requirements

ID	Requirement	Priority	Acceptance criterion	Verification
FR-01	Receive and send text messages through the Telegram Bot API	Must	A message to the test bot is stored with channel, chat id and timestamp; the bot replies within NFR-01	Integration test
FR-02	Receive and send messages through the WhatsApp Cloud API	Must (iteration 2)	Round trip with a test number; only allowed message types (templates outside the 24-hour window)	Integration test, manual
FR-03	Keep per-lead dialogue state and history across restarts	Must	After a restart the next message continues the same state	Integration test
FR-04	Run a qualification dialogue that asks only for missing fields	Must	A field already provided is asked again in at most 5% of 20 scripted scenarios	Scenario test
FR-05	Extract data into a typed lead schema and validate it	Must	100% of stored leads pass validation; invalid output is retried, then flagged for a human	Schema test
FR-06	Classify the lead as HOT, WARM or COLD with a short rationale	Must	Accuracy and macro-F1 against human labels are reported	Evaluation
FR-07	Create or update the lead in the CRM through an adapter	Must	One CRM lead per chat with fields, summary and transcript link; no duplicates on repeated runs	Integration test (mock CRM, Bitrix24 sandbox)
FR-08	Hand over to a human on triggers (explicit request, complaint, low confidence, out of scope, repeated misunderstanding)	Must	The manager is notified; the agent stops auto-replying until released; handoff recall is reported	Scenario test, evaluation
FR-09	Answer FAQ from a knowledge base without inventing prices or stock	Should	Unsupported-claim rate on the FAQ set at or below the target	Evaluation
FR-10	Operator edits fields, rules and the knowledge base without code changes	Should	A configuration change takes effect after reload	Manual test
FR-11	Log every agent decision and report metrics	Should	Each dialogue has a trace; a script prints the metrics	Inspection
FR-12	Send a follow-up if the lead is silent	Could	One reminder after a configured delay, only if channel rules allow	Manual test
FR-13	Accept a photo as a hint	Could	The photo description is added to the lead card and marked unverified	Manual test
FR-14	Instagram and TikTok channels	Won't (this release)	Documented as future work	None

6.2 Non-functional requirements

ID	Area	Priority	Acceptance criterion	Verification
NFR-01	Latency	Must	p95 first reply within 10 s (initial target)	Test run
NFR-02	Reliability	Must	No lost messages; no duplicate replies on replay	Fault test
NFR-03	Secrets	Must	No secrets in the repository; only placeholder examples	Inspection
NFR-04	Privacy	Must	Evaluation data is anonymized; retention is configurable	Inspection
NFR-05	Safety	Must	Only whitelisted tools; no destructive CRM tools; zero successful injection probes	Probe set
NFR-06	Language	Must / Should	Russian required; Kazakh and English should	Scenario test
NFR-07	Cost	Should	Cost per dialogue is reported	Evaluation
NFR-08	Maintainability	Should	Channel, LLM and CRM are replaceable through interfaces; at least 70% line coverage of core logic (initial target)	Coverage report

6.3 Lead schema (draft)

`name`, `phone`, `channel`, `intent` (buy, price, availability, consultation, other), `product_category`, `product_details` (sizes, connection type, quantity), `project_type` (apartment, house, commercial), `urgency` (today, this week, planning), `fulfilment` (delivery, pickup), `city`, `budget` (optional), `qualification` (HOT, WARM, COLD), `confidence`, `missing_fields[]`, `handoff_required`, `summary`.

6.4 Verifiability

All 14 mandatory requirements (8 functional and 6 non-functional) have a measurable criterion and a named verification method, so the fulfilment of each can be checked.

7. Architecture and diagrams

7.1 Architectural description

The system is a server-side agent. A message from Telegram or WhatsApp arrives through a webhook; the channel adapter normalizes it and a job is placed in a Redis queue so that the webhook can answer quickly. A worker loads the dialogue state and history from PostgreSQL, builds the prompt (qualification script, knowledge-base snippets, history) and calls the LLM through an interface. The model proposes the next message and the extracted fields; code, not the model, decides state transitions. Guardrails validate the output against the lead schema and the policy (no invented prices, tool whitelist). When the dialogue is complete or a handoff trigger fires, the CRM adapter creates or updates the lead in Bitrix24 and the manager is notified. Every decision is stored as a trace for evaluation.

7.2 System context diagram

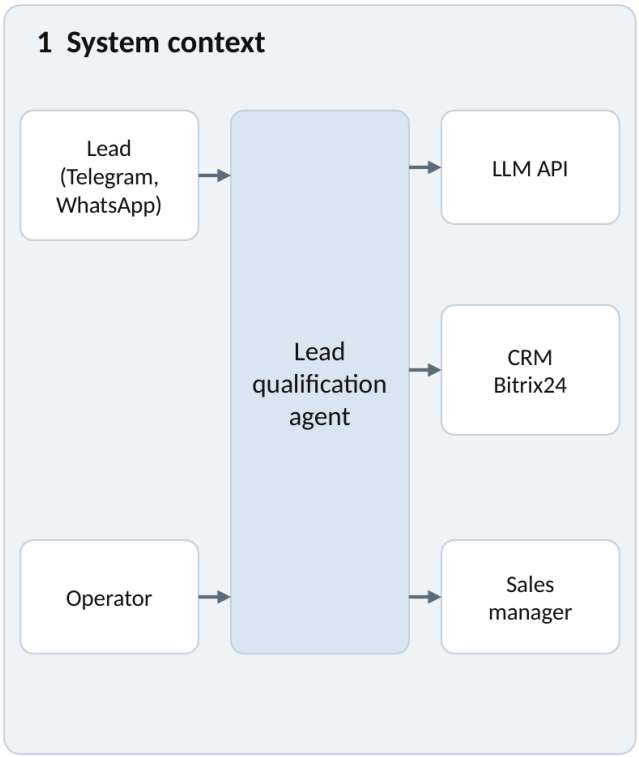


Figure 1. System context diagram

Editable source: system-context.mmd

7.3 Use case diagram

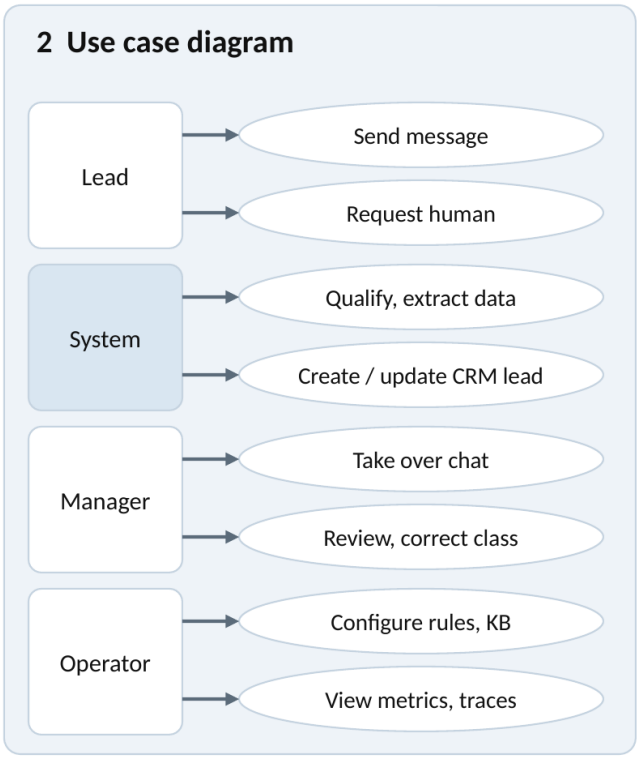


Figure 2. Use case diagram

Editable source: use-case.mmd

7.4 Component diagram (draft)

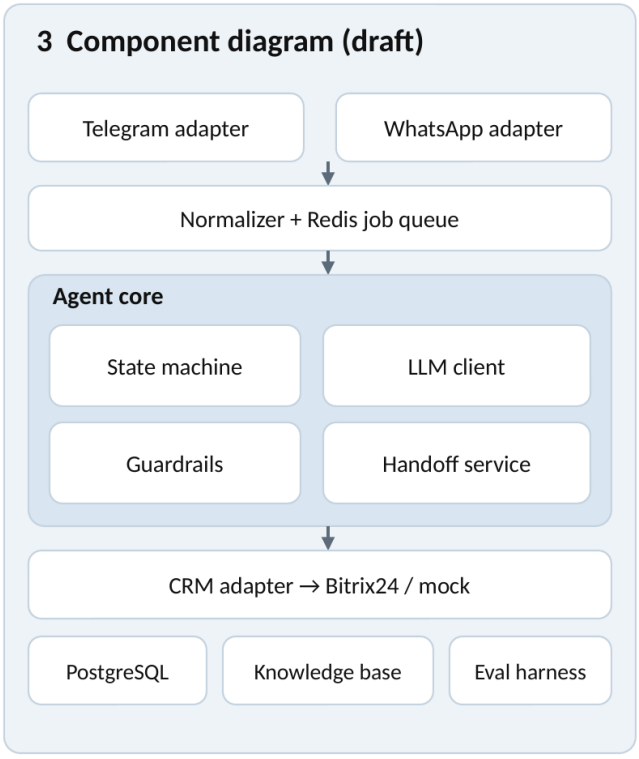


Figure 3. Component diagram (draft)

Editable source: component.mmd

7.5 Key design decisions

1. **Explicit state machine around the LLM.** States: greeting, collecting, confirming, done, plus handoff. The model suggests, code decides (literature source 5).
2. **Typed lead schema with validation.** Invalid output is retried once, then flagged for a human.
3. **Tool whitelist.** The model can call only the CRM upsert, a knowledge-base search and a handoff request; customer text is untrusted data (source 9, NFR-05).
4. **Adapter interfaces** for channel, LLM and CRM (NFR-08).
5. **Asynchronous processing** with a queue (NFR-01, NFR-02).
6. **Idempotency.** The lead key is channel plus chat id; message ids are deduplicated.

7.6 Alignment of requirements, components, data and scenarios

Requirement	Component	Data	Scenario (use case)
FR-01, FR-02	Telegram and WhatsApp adapters, normalizer, queue	Message	Send message
FR-03	Conversation manager (state machine)	Dialogue state, history	Send message
FR-04, FR-05	Prompt builder, LLM client, guardrails	Lead schema	Qualify and extract lead data
FR-06	Guardrails and classification step	Lead class and rationale	Qualify and extract lead data
FR-07	CRM adapter (Bitrix24, mock)	Lead card	Create or update CRM lead
FR-08	Handoff service	Handoff reason	Request human; take over chat
FR-09	Knowledge base	Knowledge-base items	Send message
FR-10	Configuration loader	Rules, fields	Configure rules and knowledge base
FR-11	Trace store, evaluation harness	Traces, metrics	View metrics and traces

7.7 Editable sources and exports

Diagram sources are Mermaid text files in diagrams/; PNG exports used in this report are in diagrams/png/.

8. UX / prototype

8.1 What is prototyped

The project has no end-user application of its own. People work in tools they already use: customers in a messenger, managers in the CRM, the operator in a small panel. The prototype therefore shows three key screens and the data contract between the agent and the CRM. All data on the mockups is synthetic (no real customers, no real prices); the metric tiles in the operator panel stay empty until the evaluation produces values.

8.2 Mockups of key scenarios

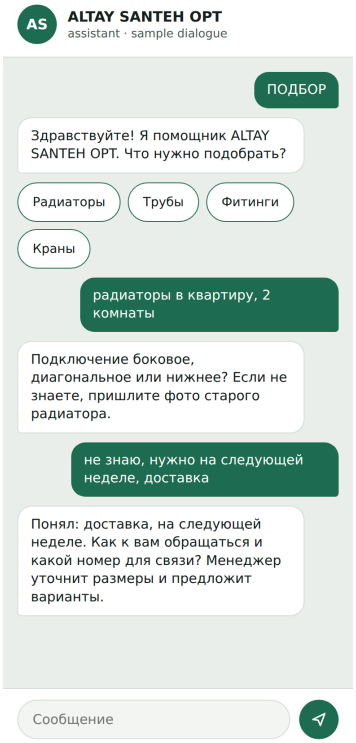


Figure 4. Customer chat with the agent (Telegram or WhatsApp)

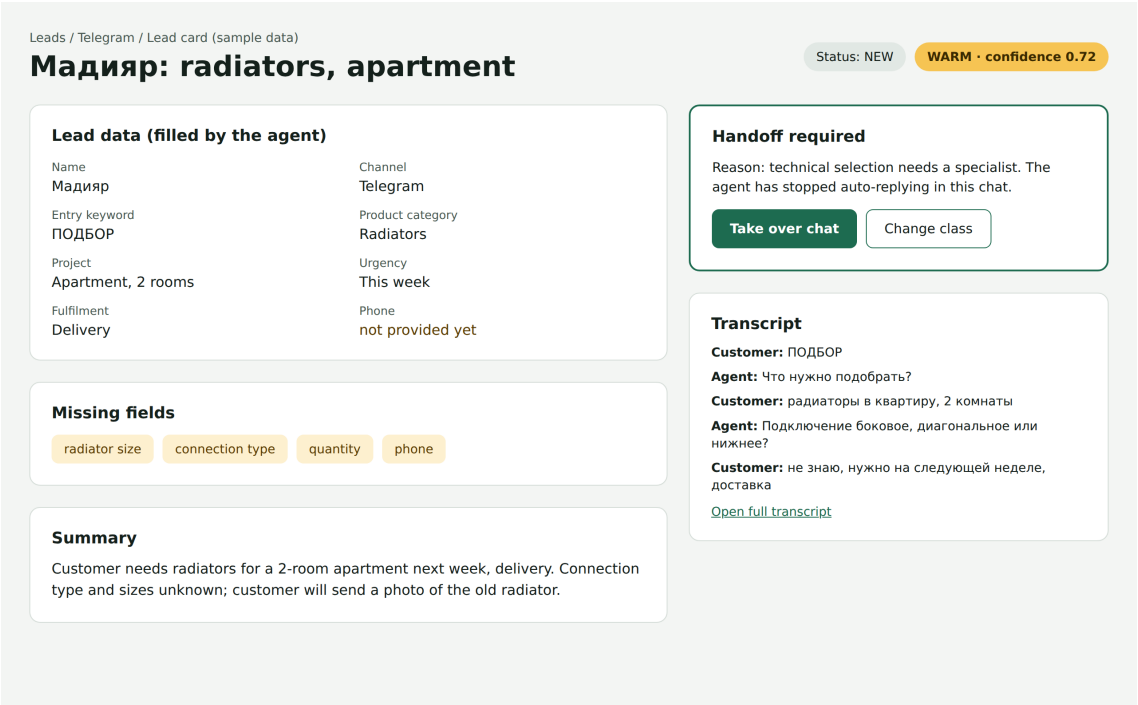


Figure 5. Lead card in the CRM as seen by the manager

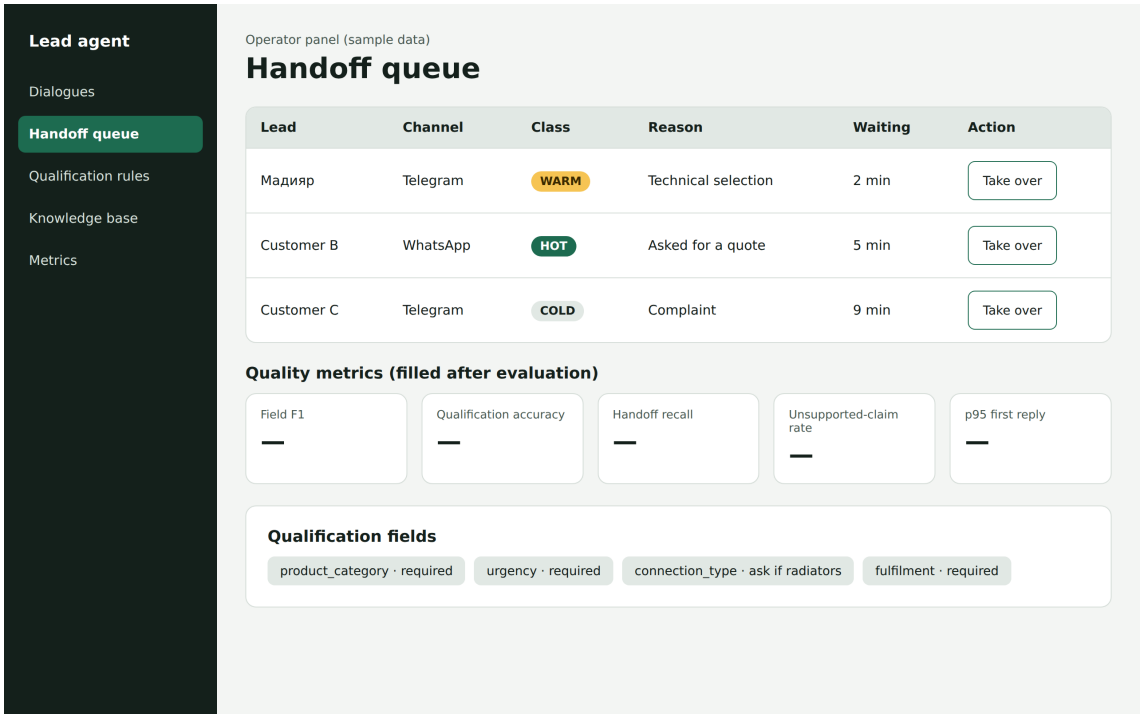


Figure 6. Operator panel: handoff queue, metrics and qualification rules

8.3 Key scenario

Step	Actor	What happens
1	Customer	Writes the keyword ПОДБОР in the messenger
2	Agent	Greets and offers the product groups (radiators, pipes, fittings, taps)
3	Customer	Answers in free text: radiators for a 2-room apartment
4	Agent	Asks only for missing fields (connection type; photo of the old radiator)
5	Customer	Gives the timing (next week) and delivery
6	Agent	Asks for name and phone, then hands over because a specialist is needed
7	System	Creates the lead card in the CRM with class WARM, missing fields and summary
8	Manager	Sees the notification, opens the chat and takes over

8.4 Data contract (output of the agent)

```
{
  "intent": "consultation",
  "product_category": "radiators",
  "project_type": "apartment",
  "urgency": "this_week",
  "fulfilment": "delivery",
  "qualification": "WARM",
  "confidence": 0.72,
  "missing_fields": ["size", "connection_type", "phone"],
  "handoff_required": true,
  "handoff_reason": "technical selection needs a specialist"
}
```

8.5 How users and external systems interact

Customers use the messenger they already have; managers work in the CRM; the operator uses the panel; external systems connect through webhooks (Telegram, WhatsApp) and the REST API (Bitrix24, LLM provider).

8.6 Prototype files

Mockup images: prototype/

9. Verification and evaluation plan

9.1 Research questions

1. How accurately does the agent extract lead fields from free-text dialogues?
2. How well does it classify HOT, WARM and COLD compared with a human manager?
3. Does it hand over to a human when it should and avoid inventing facts?
4. Is it faster than the current process, and what does it cost?

9.2 Test data

Set	Content	Size (target)	Source
T1	Anonymized real dialogues	as available, at least 30	Altay Santeh Opt, with written consent; names and phones removed
T2	Synthetic dialogues for rare cases	at least 70	Generated and reviewed by the author
T3	FAQ and no-invention probes (price, stock, discount questions)	30	Author, based on the discussions
T4	Prompt-injection probes	20	Author, informed by the OWASP LLM Top 10

Labels: two annotators (the author and, if possible, the client's manager) label fields, class and handoff; disagreement is reported as Cohen's kappa. The test set is frozen before prompt tuning; tuning uses a separate development split.

9.3 Outcome metrics

Metric	Definition	Initial target
Field F1	Per-field precision and recall against gold labels	0.85 or higher
Qualification accuracy and macro-F1	Agent class against human class	accuracy 0.80 or higher
Handoff recall and precision	Correct handoffs among required and among performed	recall 0.90 or higher
Unsupported-claim rate	Share of answers with price, stock or terms absent from the knowledge base	2% or lower
Injection success rate	Probes that cause an unauthorized tool call or policy break	0%
Task success (end state)	The CRM record equals the expected record	reported
Latency and cost	p95 first reply; cost per completed dialogue	under 10 s

Targets are hypotheses and will be calibrated with the supervisor after the baseline run.

9.4 Baselines and ablations

ID	Baseline	Purpose
B1	Rule-based decision-tree bot with fixed questions	Shows the value of free-text understanding
B2	Make / n8n with a single LLM prompt, no validation or state machine	Shows the value of the engineering layers
B3	Human manager (reply times from real chat timestamps; labels as reference)	Comparison with the real process
A1	Full system without schema validation	Ablation
A2	Free-form against constrained output format	Ablation (literature source 10)

9.5 Measurement conditions and test levels

- Fixed prompts and model version per run; temperature fixed; three runs per configuration; results as mean and range.
- Test levels: unit (parsers, schema, state machine), integration (webhook to CRM; mock CRM and a Bitrix24 sandbox), end-to-end scripted dialogues, and the evaluation harness.

9.6 How the aim will be demonstrated

The aim is demonstrated by metrics measured on labeled dialogues and compared with the three baselines. Plan: by pre-defense 1, the protocol (this section); by pre-defense 2, baselines B1 and B2 measured and the MVP evaluated on T1 and T2; at the final defense, full results, ablations and error analysis.

10. Project management and risks

10.1 Repository and tracker

- Git repository: github.com/Wolfenste1n/diploma-Madiyar, version for this report: tag prob-period-1.
- YouTrack project DIP: madiyar-diploma.youtrack.cloud.
- Conventions: branch `DIP-<n>-<slug>`, commit message `DIP-<n>: <what changed>`, one pull request per task that references the YouTrack task and states how the work was checked; the supervisor reviews significant changes at checkpoints. Tags: `prob-period-1`, then one tag per later submission.

10.2 Responsibilities

Who	Responsibility
Student (Madiyar Kenzhebayev)	Analysis, design, code, evaluation, documentation, all YouTrack tasks
Supervisor (Zhanna B. Sadirmekova)	Reviews key deliverables at agreed checkpoints
Experts and client	Discussions, approval of requirements, anonymized dialogues

10.3 Task plan (YouTrack project DIP)

ID	Task	Priority	Due date	Assignee	State
DIP-1	Create repository and README	Critical	2026-10-13	Student	see YouTrack
DIP-2	Set up YouTrack and create tasks	Critical	2026-10-13	Student	see YouTrack
DIP-3	Project proposal	Major	2026-10-13	Student	see YouTrack
DIP-4	Evidence: expert discussions and client agreement	Critical	2026-10-20	Student	see YouTrack
DIP-5	Literature review and source papers	Major	2026-10-22	Student	see YouTrack
DIP-6	Existing systems comparison	Major	2026-10-22	Student	see YouTrack
DIP-7	Technology comparison	Major	2026-10-20	Student	see YouTrack
DIP-8	Requirements specification	Critical	2026-10-20	Student	see YouTrack
DIP-9	Architecture diagrams	Major	2026-10-22	Student	see YouTrack
DIP-10	Prototype	Major	2026-10-22	Student	see YouTrack
DIP-11	Verification and evaluation plan	Major	2026-10-22	Student	see YouTrack
DIP-12	Process supervisor feedback	Normal	2026-10-24	Student	see YouTrack
DIP-13	Telegram gateway skeleton	Major	2026-11-07	Student	see YouTrack
DIP-14	Agent core: state machine and schema validation	Critical	2026-11-21	Student	see YouTrack
DIP-15	CRM adapter interface, mock CRM, Bitrix24 adapter	Major	2026-11-28	Student	see YouTrack
DIP-16	Dataset and labeling protocol	Major	2026-12-05	Student	see YouTrack
DIP-17	WhatsApp channel	Normal	2026-12-12	Student	see YouTrack
DIP-18	Baselines and evaluation harness	Major	2026-12-19	Student	see YouTrack

The current state of every task, with expected outcome, acceptance criteria and links to commits, is kept in YouTrack; the plan is also stored as DIP_task_plan.pdf.

10.4 Milestones

Milestone	Date	Content
Pre-defense 1 (probationary period 1)	24 October 2026	Analysis, requirements, architecture, prototype, evaluation plan, repository and tracker
MVP core	November 2026	Telegram gateway, agent core, mock CRM, Bitrix24 adapter
Extensions	December 2026	WhatsApp channel, injection tests
Pre-defense 2	May 2027	End-to-end MVP, baselines measured
Final defense	June 2027	Full evaluation, ablations, documentation

10.5 Risks

ID	Risk	Likelihood	Impact	Mitigation	Level
R1	The model invents prices or stock	Medium	High	Answers only from the knowledge base; handoff for quotes; unsupported-claim metric	High
R2	Prompt injection through customer text	Medium	High	Tool whitelist, no destructive CRM tools, output validation, probe set	High
R3	WhatsApp onboarding or template rules delay the channel	High	Medium	Telegram first; WhatsApp in the second iteration; aggregator as a fallback	Medium
R4	Not enough real dialogues for evaluation	Medium	High	Synthetic set reviewed by the author; limits stated honestly	Medium
R5	Personal data in dialogues	Medium	High	Anonymize before storage; written consent; configurable retention	High
R6	Client or experts unavailable	Medium	Medium	Use the second domain contact; document the limitation	Medium
R7	Scope too large for one student	Medium	High	MVP first; Instagram, photos, Kazakh stay stretch	Medium
R8	LLM cost or latency above target	Low	Medium	Shorter prompts, caching, model choice behind an interface	Low
R9	Dependence on one vendor API	Medium	Medium	LLM and CRM interfaces	Medium

10.6 Dependencies

Bitrix24 plan with REST API and webhooks; WhatsApp Business account; LLM API access and budget; written consent of the client for dialogue data; supervisor availability for checkpoints.

10.7 Next steps

Finish the remaining expert discussions and collect the client metrics; start the Telegram gateway and the agent core (DIP-13, DIP-14); keep every commit linked to a YouTrack task.

11. Supervisor feedback

The supervisor's feedback on the key deliverables and the decision on how each comment is addressed are recorded in the log below and in YouTrack task DIP-12. Each comment gets a decision and a linked change (commit or pull request), so that the review can be traced to concrete improvements.

11.1 Feedback log

Deliverable	Question or topic for review	Supervisor feedback	Decision and action	Linked change
Proposal and scope (section 1)	Telegram, WhatsApp, Bitrix24, Russian first: acceptable?			DIP-12
Evidence (section 2)	Is a third expert discussion or a client agreement required?			DIP-12
Literature and systems (sections 3, 4)	Anything to add or change?			DIP-12
Technologies (section 5)	Hosted LLM only, or also fine-tuning or an own open model?			DIP-12
Requirements and evaluation (sections 6, 9)	Are the metrics and baselines sufficient?			DIP-12
Plan (section 10)	Official pre-defense dates; document language?			DIP-12

11.2 How feedback is processed

1. The comment is entered in the log and in YouTrack (DIP-12) on the day it is received.
2. A decision is agreed with the supervisor: accept, adjust or decline with a reason.
3. The change is made on a branch, referenced in a pull request, and the commit or pull request link is added to the log.
4. The affected section of this report is updated in the next tagged version.